

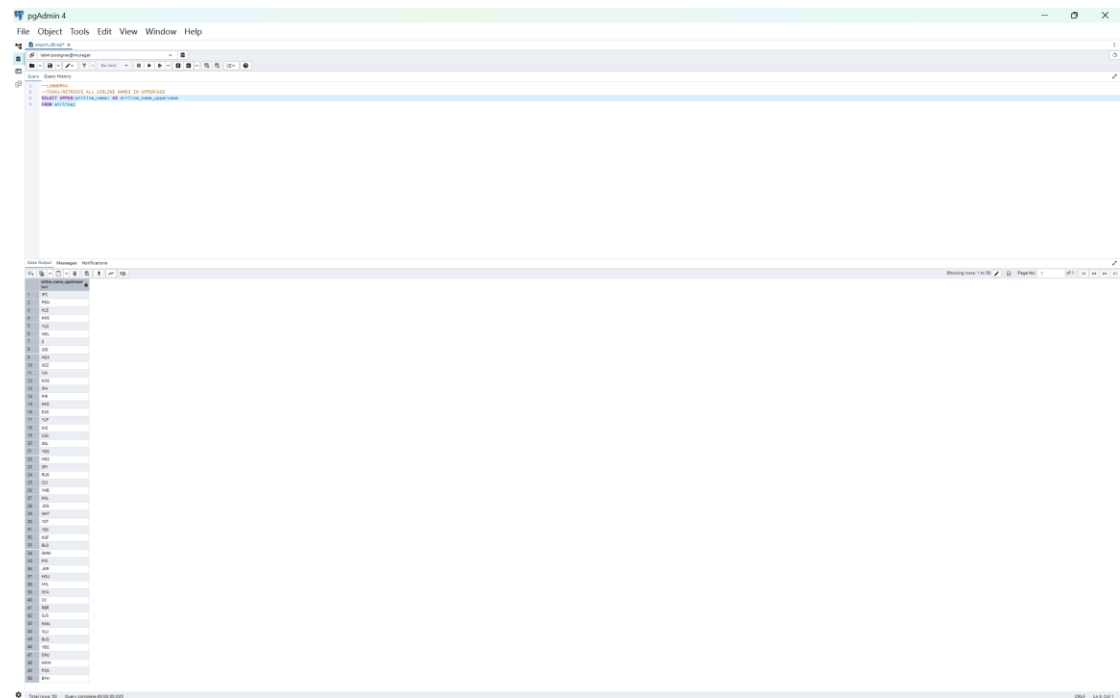
Lab4

--LABWORK4

--TASK1:RETRIEVE ALL AIRLINE NAMES IN UPPERCASE

Code:

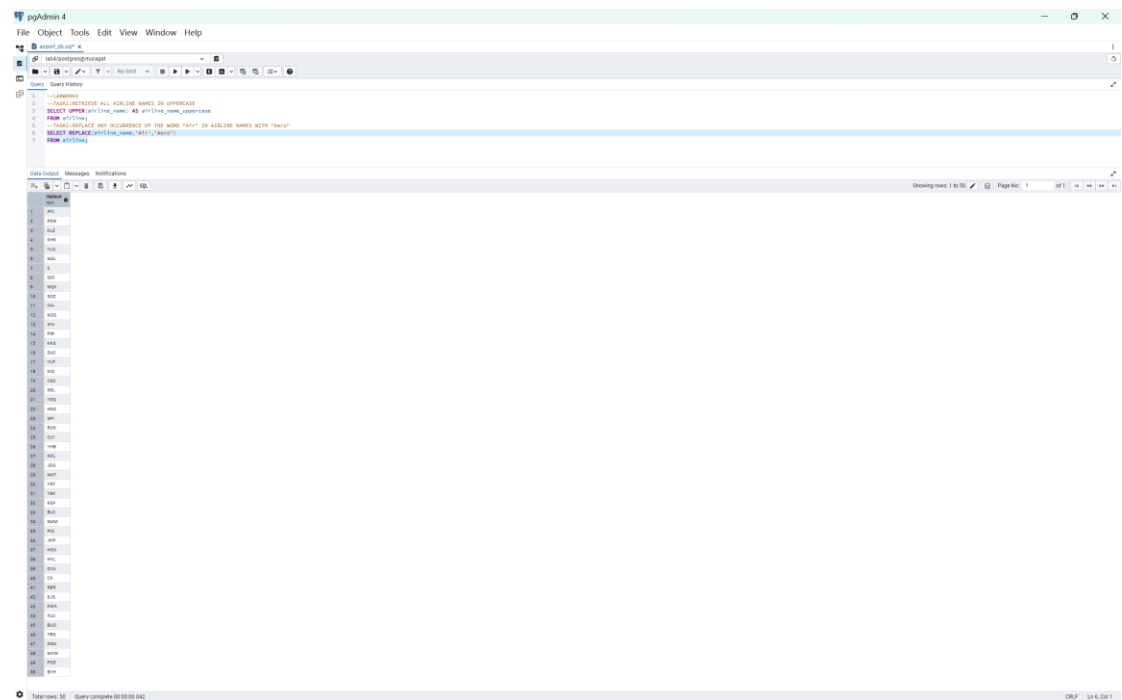
```
SELECT UPPER(airline_name) AS airline_name_uppercase  
FROM airline;
```



--TASK2:REPLACE ANY OCCURRENCE OF THE WORD "Air" IN AIRLINE NAMES WITH "Aero"

Code:

```
SELECT REPLACE(airline_name,'Air','Aero')  
FROM airline;
```



--TASK3:FIND ALL FLIGHT NUMBERS THAT COORDINATES WITH BOTH AIRLINE 1 AND AIRLINE2.

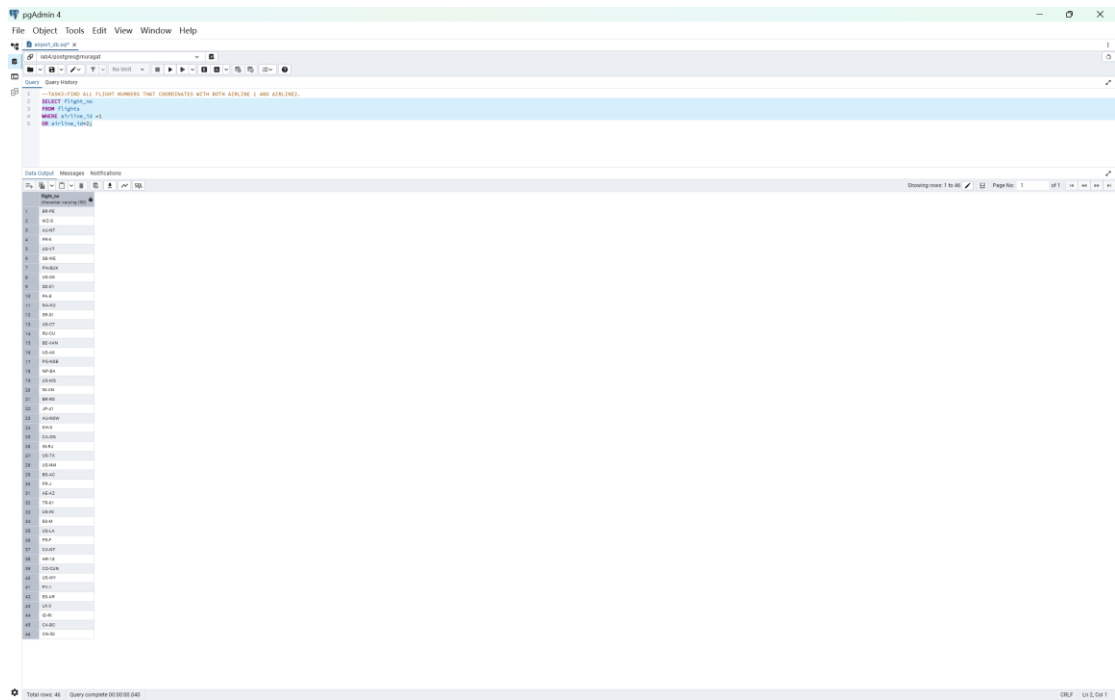
Code:

SELECT flight_no

FROM flight

WHERE airline_id = 1

OR airline_id = 2;



--TASK4: RETRIEVE AIRPORTS THAT CONTAIN THE WORD 'Reginal'AND 'Air' IN THEIR NAMES

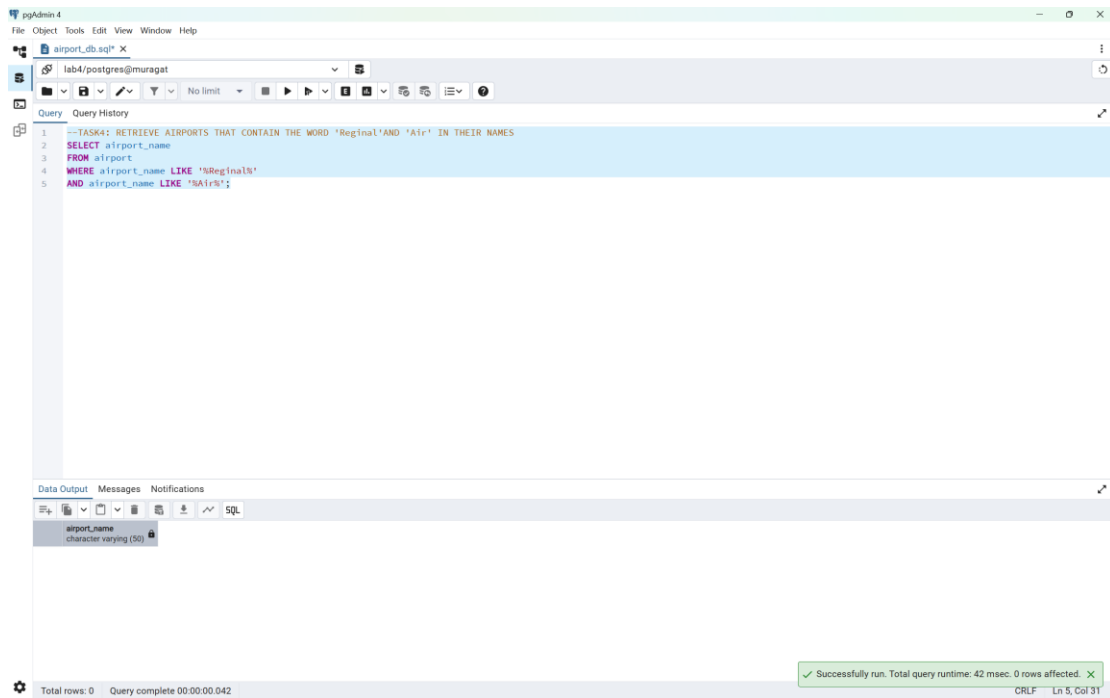
Code:

SELECT airport_name

FROM airport

WHERE airport_name LIKE '%Reginal%'

AND airport_name LIKE '%Air%';



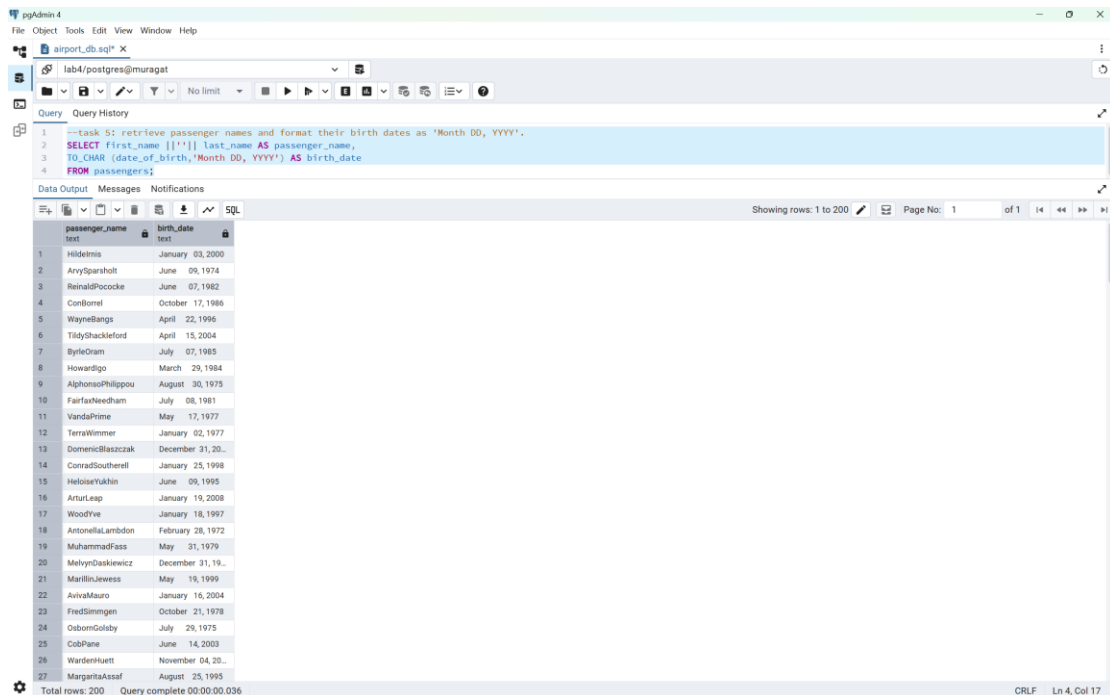
--task 5: retrieve passenger names and format their birth dates as 'Month DD, YYYY'.

Code:

SELECT first_name || ' ' || last_name AS passenger_name,

TO_CHAR (date_of_birth, 'Month DD, YYYY') AS birth_date

FROM passengers;



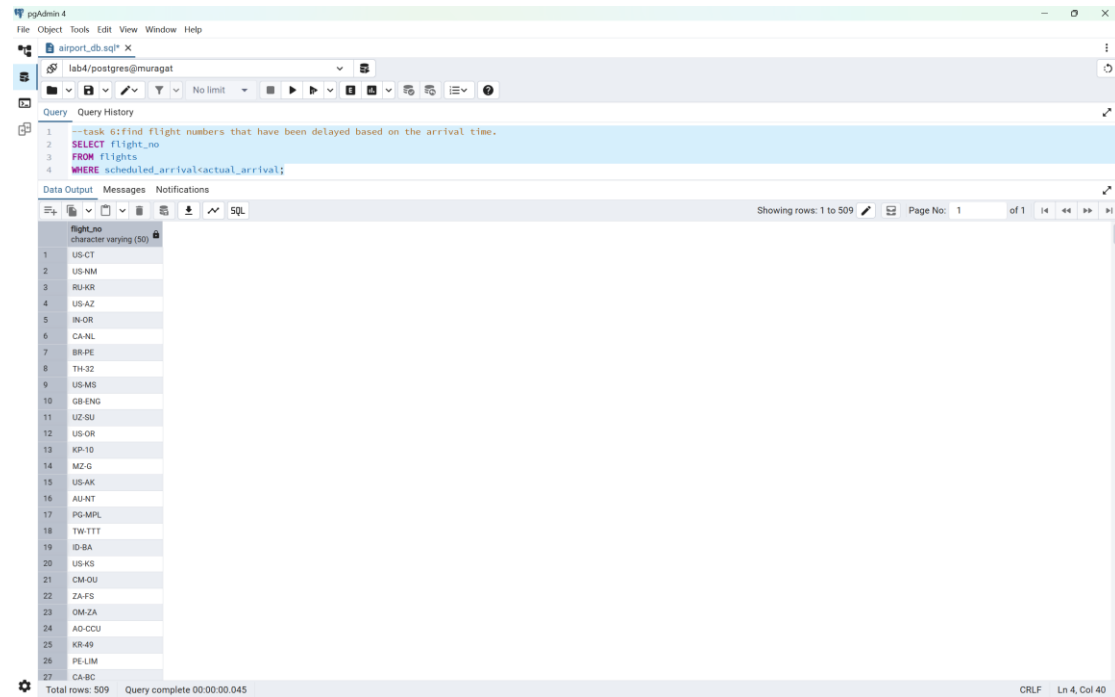
--task 6: find flight numbers that have been delayed based on the arrival time.

Code:

```
SELECT flight_no
```

```
FROM flights
```

```
WHERE scheduled_arrival<actual_arrival;
```



The screenshot shows the pgAdmin 4 interface. The query editor at the top contains the following SQL code:

```
--task 6: find flight numbers that have been delayed based on the arrival time.  
SELECT flight_no  
FROM flights  
WHERE scheduled_arrival<actual_arrival;
```

The 'Data Output' tab is active, displaying a list of flight numbers. The table has one column, 'flight_no', with a data type of 'character varying (50)'. The results are as follows:

flight_no
US-CT
US-NM
RU-KR
US-AZ
IN-OR
CA-NL
BR-PE
TH-32
US-MS
GB-ENG
UZ-SU
US-OR
KP-10
MZ-G
US-AK
AU-NT
PG-MPL
TW-TTT
ID-BA
US-KS
CM-OU
ZA-FS
OM-ZA
AO-CCU
KR-49
PE-LIM
CA-BC

The status bar at the bottom indicates 'Total rows: 509' and 'Query complete 00:00:00.045'.

--task 7: find flights that arrived late according to their actual arrival time compared to the scheduled arrival time.

Code:

```
SELECT *
```

```
FROM flights
```

```
WHERE scheduled_arrival<actual_arrival;
```

pgAdmin 4

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Query Query History

```

1 --task 7: find flights that arrived late according to their actual arrival time compared to the scheduled arrival time.
2 SELECT *
3 FROM flights
4 WHERE scheduled_arrival < actual_arrival;

```

Data Output Messages Notifications

Showing rows: 1 to 509 Page No: 1 of 1 14 44 99 91

	flight_no	scheduled_departure	scheduled_arrival	departure_airport_id	arrival_airport_id	departing_gate	arriving_gate	airline_id	status	actual_departure	actual_arrival	created_at	update_at
	character varying (50)	date	date	integer	integer	character varying (50)	character varying (50)	integer	character varying (50)	date	date	date	date
6	CA-NL	2023-03-22	2023-06-04	12	13	40	2592	27	Boarding	2023-04-28	2023-11-17	2023-11-29	2023-04-25
7	BR-PE	2024-01-16	2023-06-02	13	20	1891	5429	1	Delayed	2023-08-19	2023-11-09	2023-06-29	2023-11-09
8	TH-32	2024-02-05	2023-03-30	8	19	478	[null]	17	Boarding	2023-04-21	2023-07-21	2023-10-12	2024-03-06
9	US-MS	2024-02-06	2024-01-22	10	1	346	56	38	Delayed	2023-10-30	2024-03-06	2023-07-26	2023-09-17
10	GB-ENG	2023-11-03	2023-10-11	2	2	400	85	44	Delayed	2024-02-05	2024-01-04	2023-11-09	2023-08-12
11	UZ-SU	2023-05-13	2023-05-24	11	2	1027	190	43	Delayed	2024-02-24	2023-12-23	2023-07-21	2023-11-06
12	US-OR	2024-03-10	2023-07-14	6	14	214	285	11	Departed	2023-09-20	2023-10-23	2023-12-11	2023-07-06
13	KP-10	2024-03-12	2023-06-06	2	2	4547	2999	25	Departed	2023-03-28	2023-10-01	2023-08-21	2023-03-27
14	MZ-G	2023-09-21	2023-11-29	4	19	[null]	9380	1	Delayed	2023-04-14	2024-01-16	2024-01-04	2023-06-07
15	US-AK	2024-02-08	2023-10-20	11	3	38	13	47	Departed	2023-12-09	2024-02-21	2023-07-22	2024-01-27
16	AU-NT	2023-03-29	2023-05-06	12	14	[null]	13	1	Delayed	2023-12-30	2023-08-21	2023-07-27	2023-12-27
17	PG-MPL	2023-11-23	2023-06-05	12	12	2950	505	42	Boarding	2024-01-14	2023-07-04	2023-05-15	2023-07-24
18	TW-TTT	2023-07-04	2023-06-02	1	17	28	29	49	Delayed	2023-11-09	2023-11-14	2023-07-24	2023-04-06
19	ID-BA	2023-12-30	2023-04-04	2	9	14	5586	16	Boarding	2023-05-10	2023-10-05	2023-09-24	2023-08-27
20	US-KS	2023-03-25	2023-09-01	17	6	1691	973	28	Delayed	2024-02-01	2023-10-11	2024-01-23	2024-02-02
21	CM-OU	2023-12-01	2023-03-21	10	16	3963	10	20	Delayed	2023-05-14	2023-04-10	2024-02-07	2023-06-02
22	ZA-FS	2023-05-08	2023-06-05	8	2	5315	60	10	Delayed	2023-10-24	2023-11-11	2024-02-21	2024-03-10
23	OM-ZA	2023-11-22	2023-07-11	12	17	552	194	36	Boarding	2023-09-01	2023-11-03	2023-12-28	2023-04-14
24	AO-CCU	2023-09-01	2023-04-07	20	2	[null]	732	9	Delayed	2023-11-02	2024-03-17	2023-08-13	2023-03-20
25	KR-49	2024-02-20	2024-01-11	13	10	1171	20	23	Boarding	2023-06-08	2024-02-27	2023-06-18	2024-03-14
26	PE-LIM	2024-02-23	2023-07-02	14	5	769	99	29	Delayed	2023-05-04	2023-11-10	2023-03-24	2024-01-10
27	CA-BC	2023-11-04	2023-05-11	10	7	0	52	13	Boarding	2023-07-10	2023-08-10	2023-05-16	2023-09-23
28	SD-24	2023-04-28	2023-06-01	7	17	1582	49	37	Boarding	2023-04-28	2023-11-09	2023-08-25	2023-07-04
29	US-OH	2023-04-10	2023-09-17	8	14	993	3700	37	Delayed	2024-03-09	2024-01-26	2023-07-14	2023-06-19
30	CY-02	2023-09-14	2023-05-08	16	18	404	207	9	Delayed	2024-01-14	2024-01-25	2023-08-24	2023-08-17

Total rows: 509 Query complete 00:00:00.095 CRLF Ln 4, Col 40

--task 8:show airlines from any of:('France','Portugal','Poland') created between '2023-11-01'and'2024-03-31'.

Code:

SELECT *

FROM airline

WHERE airline_country IN ('France','Portugal','Poland')

AND created_at '2023-11-01' AND '2024-03-31';

pgAdmin 4

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airport_db.sql X

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Query Query History

```

1 --task 8: show airlines from any of ('France','Portugal','Poland') created between '2023-11-01' and '2024-03-31'.
2 SELECT *
3 FROM airline
4 WHERE airline_country IN ('France','Portugal','Poland')
5 AND created_at BETWEEN '2023-11-01' AND '2024-03-31';

```

Data Output Messages Notifications

Showing rows: 1 to 7 Page No: 1 of 1

	airline_id [PK] integer	airline_code character varying (50)	airline_name character varying (50)	airline_country character varying (50)	created_at date	update_at date
1	4	OOKB	KHS	France	2023-11-18	2024-02-18
2	13	OIFE	IFH	Portugal	2024-01-09	2023-11-09
3	15	SYKZ	KKG	Poland	2023-12-24	2023-10-30
4	24	ZSRG	RUG	Portugal	2024-03-11	2023-08-24
5	28	RKPD	JDG	Poland	2023-11-03	2023-12-06
6	30	CYST	YST	Poland	2023-11-24	2023-10-07
7	43	AYKM	KMA	France	2023-11-21	2023-05-30

Total rows: 7 Query complete 00:00:00.054

CRLF Ln 5, Col 54

--task9: find top3 overweighted baggage with more than 25kg

Code:

SELECT *

FROM baggage

WHERE weight_in_kg >25

LIMIT 3;

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Query Query History

```

1 --task9: find top3 overweighted baggage with more than 25kg
2 SELECT *
3 FROM baggage
4 WHERE weight_in_kg >25
5 LIMIT 3;

```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	baggage_id [PK] integer	weight_in_kg numeric (4,2)	created_date date	update_date date	booking_id integer
1	2	42.74	2023-12-02	2023-10-29	115
2	5	32.65	2023-05-17	2023-07-21	363
3	7	34.94	2023-04-01	2023-04-07	494

Total rows: 3 Query complete 00:00:00.164

CRLF Ln 5, Col 9

--TASK10:FIND THE YOUNGEST PASSENGER FULL NAME

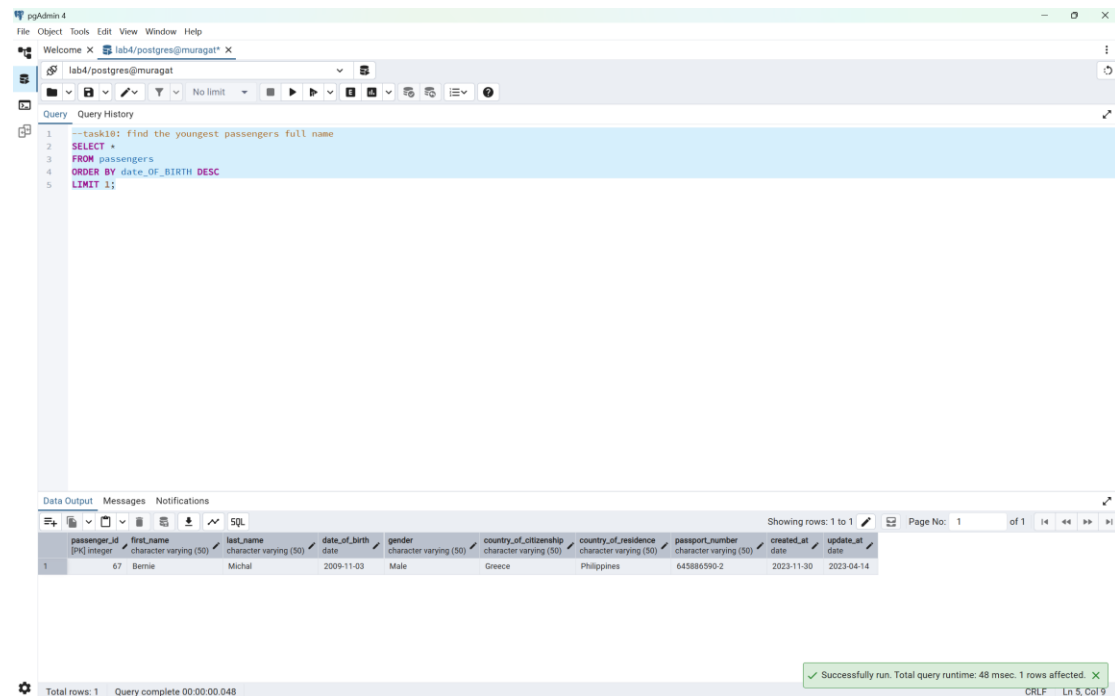
CODE:

SELECT *

FROM passengers

GROUP BY data_OF_BIRTH DESC

LIMIT 1;



The screenshot shows the pgAdmin 4 web interface. The query editor at the top contains the following SQL code:

```
--task10: find the youngest passengers full name
SELECT *
FROM passengers
ORDER BY date_OF_BIRTH DESC
LIMIT 1;
```

Below the query editor, the 'Data Output' tab is active, displaying a single row of data from the 'passengers' table. The columns and their values are:

passenger_id	first_name	last_name	date_of_birth	gender	country_of_citizenship	country_of_residence	passport_number	created_at	update_at
67	Bernie	Michal	2009-11-03	Male	Greece	Philippines	645886590-2	2023-11-30	2023-04-14

At the bottom of the interface, a status bar indicates: 'Total rows: 1 Query complete 00:00:00.048'. A green message box at the bottom right states: 'Successfully run. Total query runtime: 48 msec. 1 rows affected.'

TASK11: FIND THE CHEAPEST TICKET PRICE IN EACH BOOKING PLATFORM

SELECT booking_platform

MIN(price) AS booking_price

FROM booking

GROUP BY booking_platform;

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Query Query History

```

1 --TASK11:FIND THE CHEAPEST BOOKING PRICE ON EACH BOOKING PLATFORM
2 SELECT booking_platform,
3 MIN (price) AS minimum_price
4 FROM booking
5 GROUP BY booking_platform;
6

```

Data Output Messages Notifications

Showing rows: 1 to 494 Page No: 1 of 1

	booking_platform character varying (50)	minimum_price numeric
1	Schmitt-Gugley	4966.58
2	Rohan, Hamill and Bosco	7118.84
3	Kutch-Russel	2113.46
4	Vreun-Corwin	2183.14
5	Wofft, Wuckert and Satterfield	4911.89
6	Reichel, Deckow and Quitzon	8389.34
7	Murazik-Gulgowski	950.80
8	Cole, Corwin and Leannon	8735.70
9	Franecki, Dickens and Walsh	283.78
10	McLaughlin, Sipes and Abernathy	7490.57
11	Leffler-Thompson	2559.77
12	Raynor Inc	410.47
13	Waelchi Inc	3684.97
14	Renner, Weber and Wilkinson	4230.67
15	Langworth-Osinski	4982.91
16	Will LLC	6462.07
17	Stroman, Block and Halvorson	938.92
18	Mraz, Daniel and Schmeler	2077.18
19	Wisozk and Sons	868.47
20	Wuckert, Weber and Walter	2520.04
21	Treutel, Zemlak and Jast	7379.88
22	Graham, Torp and Mante	7813.55
23	Kahlerin Group	3553.63

Total rows: 494 Query complete 00:00:00.047 CRLF Ln 6, Col 1

--task12:return airlines whose airline_code contains a digit

Code:

SELECT *

FROM airline

WHERE airline_code ~'[0-9]';

pgAdmin 4

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Query Query History

```

1 --task12:return airlines whose airline_code contains a digit
2 SELECT *
3 FROM airline
4 WHERE airline_code ~'[0-9]';

```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

airline_id [PK] integer	airline_code character varying (50)	airline_name character varying (50)	airline_country character varying (50)	created_at date	update_at date
1	22 2AK6	HGZ	Philippines	2024-03-13	2023-10-20

Total rows: 1 Query complete 00:00:00.050 CRLF Ln 4, Col 29

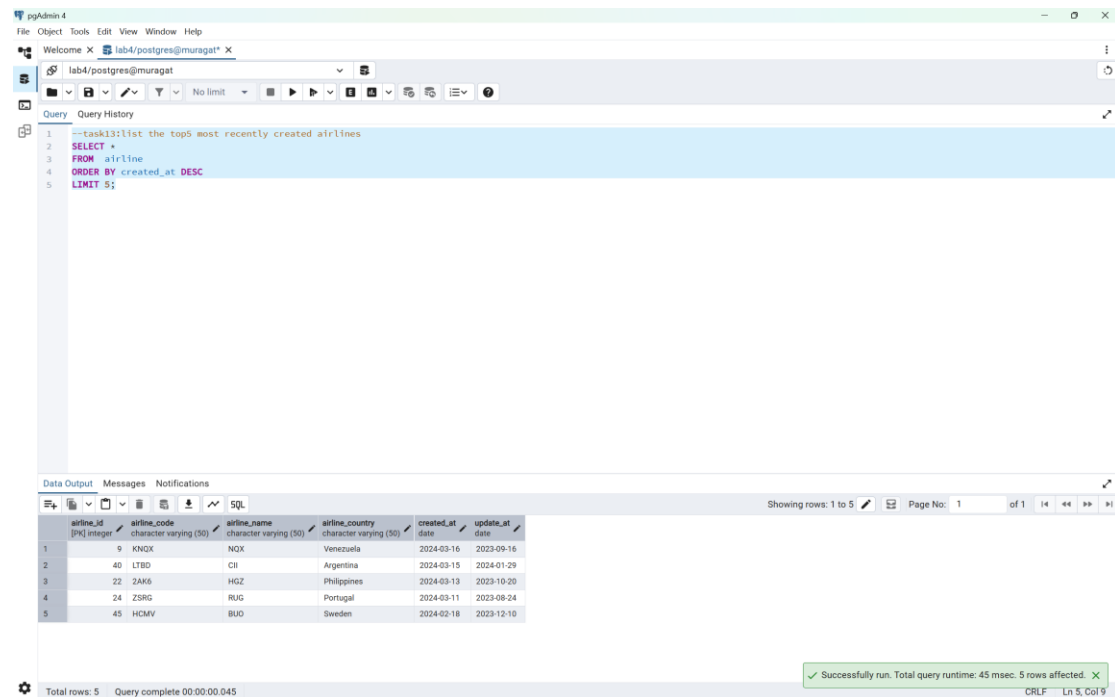
--task13: list the top5 recently created airlines

SELECT *

FROM airline

ORDER BY created_at DESC

LIMIT 5;



The screenshot shows the pgAdmin 4 interface. The SQL query editor contains the following query:

```
--task13: list the top5 most recently created airlines
SELECT *
FROM airline
ORDER BY created_at DESC
LIMIT 5;
```

The query results are displayed in a table with the following columns: airline_id [PK] integer, airline_code character varying (50), airline_name character varying (50), airline_country character varying (50), created_at date, and update_at date. The results show the top 5 airlines by creation date.

airline_id [PK] integer	airline_code character varying (50)	airline_name character varying (50)	airline_country character varying (50)	created_at date	update_at date
9	KNQX	NQX	Venezuela	2024-03-16	2023-09-16
40	LTBD	CII	Argentina	2024-03-15	2024-01-29
22	2AK6	HGZ	Philippines	2024-03-13	2023-10-20
24	ZSRG	RUG	Portugal	2024-03-11	2023-08-24
45	HCMV	BUO	Sweden	2024-02-18	2023-12-10

The status bar at the bottom indicates: Total rows: 5, Query complete 00:00:00.045, Successfully run. Total query runtime: 45 msec. 5 rows affected.

---task14: Return all rows where booking_id is between 200 and 300 inclusive and check_result <> 'Checked'.

Code:

SELECT *

FROM public.baggage_check

WHERE booking_id BETWEEN 200 AND 300

AND CHECK_RESULT <> 'CHECKED';

The screenshot shows the pgAdmin 4 interface. The query editor at the top contains the following SQL query:

```

1  --task14: Return all rows where booking_id is between 200 and 300 inclusive and check_result <> 'Checked'.
2
3  SELECT *
4  FROM public.baggage_check
5  WHERE booking_id BETWEEN 200 AND 300
6  AND check_result <> 'Checked';

```

The query results are displayed in a table with the following columns: baggage_check_id, check_result, created_at, update_at, booking_id, and passenger_id. The table shows 26 rows of data, all with 'Not checked' as the check_result. The status bar at the bottom indicates 'Total rows: 56' and 'Query complete 00:00:00.043'.

baggage_check_id	check_result	created_at	update_at	booking_id	passenger_id
7	Not checked	2023-05-04	2023-04-29	226	44
12	Not checked	2023-10-02	2023-06-22	288	133
22	Not checked	2023-06-14	2023-11-10	246	101
23	Not checked	2023-05-29	2023-06-02	284	22
24	Not checked	2024-02-17	2023-10-03	267	68
25	Not checked	2024-01-20	2023-06-10	237	140
61	Not checked	2023-06-12	2023-11-05	279	195
73	Not checked	2023-12-02	2023-05-28	274	187
75	Not checked	2023-11-03	2023-12-03	300	130
83	Not checked	2023-10-04	2023-04-26	259	63
91	Not checked	2023-05-21	2023-09-03	234	76
92	Not checked	2024-01-26	2023-04-07	260	154
107	Not checked	2024-03-06	2023-05-01	218	188
110	Not checked	2023-07-16	2024-02-09	256	109
124	Not checked	2023-09-10	2024-01-09	226	24
152	Not checked	2023-08-23	2023-07-25	250	154
158	Not checked	2024-01-26	2023-07-05	245	33
200	Not checked	2024-01-22	2023-08-27	231	174
212	Not checked	2023-06-06	2023-12-25	246	7
213	Not checked	2023-08-07	2024-01-05	237	25
224	Not checked	2023-10-04	2023-09-22	263	123
239	Not checked	2023-10-23	2023-05-09	225	173
240	Not checked	2024-02-07	2024-01-02	264	199
244	Not checked	2023-07-12	2023-05-16	294	190
246	Not checked	2023-10-10	2023-04-07	238	150
251	Not checked	2023-06-19	2023-06-16	260	26

--TASK15:Baggage checks where update_at is in the same month as created_at but occurs earlier than created_at.

Code:

```
SELECT *
```

```
FROM public.baggage_check
```

```
WHERE EXTRACT(YEAR FROM update_at) = EXTRACT(YEAR FROM created_at)
```

```
AND EXTRACT(MONTH FROM update_at) = EXTRACT(MONTH FROM created_at)
```

```
AND update_at < created_at;
```

pgAdmin 4

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No limit

Query Query History

```
1 --TASK15:Baggage checks where update_at is in the same month as created_at but occurs earlier than created_at.
2 SELECT *
3 FROM public.baggage_check
4 WHERE EXTRACT(YEAR FROM update_at) = EXTRACT(YEAR FROM created_at)
5 AND EXTRACT(MONTH FROM update_at) = EXTRACT(MONTH FROM created_at)
6 AND update_at < created_at;
```

Data Output Messages Notifications

Showing rows: 1 to 18 SQL

	baggage_check_id [PK] integer	check_result character varying (50)	created_at date	update_at date	booking_id integer	passenger_id integer
1	34	Checked	2023-04-25	2023-04-08	28	57
2	86	Checked	2023-12-25	2023-12-07	312	22
3	104	Checked	2024-03-17	2024-03-14	41	83
4	109	Not checked	2023-08-30	2023-08-09	469	91
5	119	Not checked	2023-12-02	2023-12-01	135	55
6	148	Checked	2024-03-17	2024-03-16	86	5
7	219	Checked	2023-04-17	2023-04-14	233	200
8	248	Not checked	2023-06-22	2023-06-09	449	60
9	251	Not checked	2023-06-19	2023-06-16	260	26
10	299	Checked	2023-04-26	2023-04-08	385	191
11	304	Not checked	2023-07-29	2023-07-22	210	10
12	396	Not checked	2023-07-13	2023-07-01	37	144
13	412	Not checked	2023-09-30	2023-09-27	347	70
14	423	Not checked	2024-02-22	2024-02-10	449	148
15	444	Not checked	2023-05-18	2023-05-02	282	59
16	455	Not checked	2023-12-31	2023-12-05	321	159
17	474	Not checked	2024-01-28	2024-01-19	75	64
18	494	Checked	2023-06-13	2023-06-09	477	18

Total rows: 18 Query complete 00:00:00.054 CRLF Ln 6, Col 28